

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2020

ESE 4115

(Power Plant Engineering)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Classify power plants. What are the major sources of energy for power generation? 06
- 1(b). What are the effects of variable load on power plant design and operation? 08
- 1(c). What are factors to be considered while selecting site for a power plant? 08
- 1(d). A power station is said to have a use factor of 45% and capacity factor of 35%. For how many hours in a year was the power station not in service? 08
- 2(a). What is meant by depreciation? What factors influence the depreciation? 06
- 2(b). Describe the importance of load forecasting. 11
- 2(c). Calculate the generating cost per kWh delivered from a generating station from the following data: 13
Plant capacity = 100 MW, Annual load factor = 45%
Capital cost = Tk 150 crores
Annual cost of wages, taxation etc. = Tk 1 crores
Cost of fuel, lubrication, maintenance = Tk 0.5/kWh generated
Interest = 6% per annum, Depreciation = 8% per annum of initial cost
- 3(a). What are the factors to be considered while selecting turbines for hydroelectric power plant? 05
- 3(b). Why is hydropower called renewable source of energy? 06
- 3(c). Why governing of hydroelectric power plant is required? 04
- 3(d). The following data refers to a proposed hydroelectric power plant: 15
Catchment area = 800 km², Rainfall = 180 cm/year
Available head = 32 m, Infiltration and evaporation loss = 40%
Turbine efficiency = 85%, Generator efficiency = 90%
Load factor = 0.5
(i) Calculate power developed (ii) Suggest suitable turbines for the plant

SECTION – B

- 4(a). Is diesel power plant suitable as peak load plant? Describe briefly. 08
- 4(b). What are main disadvantages of diesel engine power plant? Explain the cooling system of diesel engine power plant. 12
- 4(c). Describe the methods of starting of a diesel power plant. 10
- 5(a). Is it possible to increase the efficiency of gas turbine power plant by using exhaust gas? Describe. 16
- 5(b). A gas turbine power plant operating on an ideal Brayton cycle has a pressure ratio of 9. The gas temperature is 310K at the compressor inlet and 1100K at the turbine inlet. For the air standard assumptions, calculate the thermal efficiency. 14
- 6(a). Make a comparison between natural and mechanical draught systems. 08
- 6(b). Describe the various factors to be considered while selecting the site for steam power plant. 13
- 6(c). Determine the height of chimney to produce a static draught of 22 mm of water if the mean flue gas temperature in a chimney is 290°C and ambient temperature in boiler house is 20°C. The gas constant (R) for air is 29.26 kgf·m/kg·K and for chimney flue gas is 26.2 kgf·m/kg·K. Assume barometer reading as 760 mm of mercury. 09

--- END ---

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY**Department of Energy Science and Engineering**B. Sc. Engineering 4th Year 1st Term Examination, 2020

ESE 4025

(HVAC&R System Design)

Time: 1 Hour 30 Minutes.

Full Marks: 120

- N.B. i) Answer any TWO questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is meant by TR? Write down the differences between refrigeration and air conditioning system. 08
- 1(b). Discuss about the thermodynamic process of theoretical vapor compression refrigeration cycle with dry saturated vapor after compression with necessary diagrams. 12
- 1(c). 600 kg of fruits are supplied to a cold storage at 25°C. The cold storage is maintained at –5°C and the fruits get cooled to the storage temperature in 8 hours. The latent heat of freezing is 105 kJ/kg and specific heat of fruit is 1.255 kJ/kgK. Find the refrigeration capacity of the plant. 10
- 2(a). Explain the environmental and safety properties of refrigerant. 07
- 2(b). Define multistage compression. Explain the single compressor-multiple expansion valves with the required p-h diagram. 11
- 2(c). A standard VCR system developing 50 kW of refrigeration using R22 refrigerant operates with a condensing temperature of 35°C and an evaporating temperature of –10°C. The enthalpy of the refrigerant after the compression is 435.2 kJ/kg. The required values for R22 refrigerant are given in table. By using these values, determine: (i) The refrigeration effect in kJ/kg; (ii) The circulation rate of refrigerant in kg/s; (iii) The power required by the compressor in kW; and (iv) COP. 12

Table: Properties of saturated liquid and saturated vapor of R22

Temperature (°C)	Pressure (bar)	Enthalpy (kJ/kg)		Entropy (kJ/kgK)	
		Liquid	Vapor	Liquid	Vapor
–12	3.3040	186.09	400.27	0.9484	1.7686
–8	3.8059	190.69	401.85	0.9657	1.7621
34	13.215	241.84	414.79	1.1419	1.7050
36	13.898	244.44	415.18	1.1501	1.7024

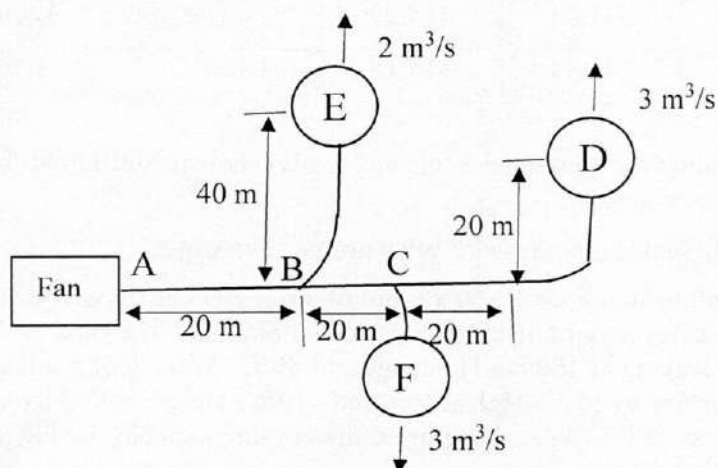
- 3(a). Draw double effect absorption refrigeration system and explain how it is different from triple effect absorption refrigeration system. 10
- 3(b). What is meant by hermetic sealed compressor? What are its advantages? 07
- 3(c). In aircraft refrigeration unit using air cycle, 50 kg/min of air at 200 cm Hg gauge and 205°C are bled off the air compressor serving the jet engine of an airplane. The bled air is passed through a heat exchanger leaving at 180 cm Hg gauge and 80°C. At this point it is expanded through a small cooling turbine to 20 cm Hg vacuum and –10°C. The air exhausted out of the plane is at 25°C. Assume $c_p=1.0$ kJ/kgK. Determine the cooling capacity in TR, and COP based on input power to bled off air. 13

SECTION – B

- | | | |
|-------|--|----|
| 4(a). | What is meant by bypass factor? How do you draw sensible heat factor (SHF) line on a psychrometric chart? | 10 |
| 4(b). | What is meant by thermal comfort? What are the factors affecting thermal comfort? | 10 |
| 4(c). | Distinguish between effective temperature and operative temperature. | 06 |
| 4(d). | Draw the ASHRAE comfort chart. | 04 |
| | | |
| 5(a). | Classify air-conditioning system. Explain the working principle of split air-conditioning system with schematic diagram. | 09 |
| 5(b). | Distinguish between constant air volume and variable air volume of all-air central air-conditioning system. What are the advantages of all-air systems? | 07 |
| 5(c). | What are the effects of indoor air quality on health and comfort? How to control indoor air quality? | 06 |
| 5(d). | What are the advantages of VRF system? Draw the block diagram of two-pipe heat recovery VRF system. | 08 |
| | | |
| 6(a). | What is cooling load? List the main components of cooling load estimation. | 10 |
| 6(b). | What are the general rules of duct design? | 08 |
| 6(c). | Determine the dimensions of AB, BC, CD as shown in figure using equal friction method. Chose a friction rate of 0.09 mmH ₂ O/m length of the duct. Use the following formula for friction rate: | 12 |

$$\frac{\Delta p_f}{L} = 0.002268 \times \frac{\dot{Q}_v^{1.852}}{D^{4.973}} \text{ mm H}_2\text{O/m}$$

Where $\dot{Q}_v$ is in m³/s and D is in m.



KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2020

ESE 4105

(Fuel Combustion and IC Engines)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | |
|--|----|
| 1(a). How can you use hydrogen as a fuel in CI Engine? Explain the physical significances of octane number and cetane number with example. | 11 |
| 1(b). What are the problems of methanol as alternative fuel? Why do you need extra fuel when the throttle is opened? | 14 |
| 1(c). Why are the intake valves larger than exhaust valves? | 05 |
| 2(a). What is catalytic converter? With a schematic diagram explain how a 3-way catalytic converter does treat exhaust from engine? | 12 |
| 2(b). “Four stroke IC engine is economical and less pollutant than two stroke engine”- Justify the statement. | 10 |
| 2(c). How is the circulation achieved in thermo-syphon system? What are the limitations of this system? | 08 |
| 3(a). What is meant by total emission control packages? How NO _x and PM are generated in CI engines? | 10 |
| 3(b). What is blow-by loss? How can you achieve effective removal of blow by? Explain with schematic diagram. | 12 |
| 3(c). What is smoke? “ Smoke colors are indicative of dominant source of particulate”- Explain. | 08 |

SECTION – B

- 4(a). Discuss the following engine performance parameter with necessary diagrams: 09
- (i) Brake power
 - (ii) Specific fuel consumption
 - (iii) Volumetric efficiency
- 4(b). What is meant by fuel rich combustion and fuel lean combustion? Why two stroke CI engines find wider use whereas two stroke SI engines are used only to a limited extent? 09
- 4(c). What is meant by “wide open throttle (WOT)” operation? Why pumping loss is higher in conventional SI engine compared to CI engine? How to reduce it? 12
- 5(a). How flames are classified? Describe the different stages of combustion in CI engine with the help of “rate of heat release-crank angle” diagram. 10
- 5(b). How ignition delay in SI engine differs from ignition delay in CI engine? “Factors tend to increase knock in SI engine reduce knock in CI engine”- Justify the above statement. 12
- 5(c). What is meant by abnormal combustion? Explain the origin of knock in SI engine with the help of auto-ignition theory. 08
- 6(a). Discuss the supercharging limits for both SI engine and CI engine. 10
- 6(b). Describe the effects of supercharging. Which engine is more suitable for supercharging- SI engine or CI engine? Why? 10
- 6(c). How can you modify an engine for supercharging? What are the limitations of turbocharging? 10

--- END ---

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2020

ESE 4007

(Smart Grid Technology)

Time: 1 Hour 30 Minutes.

Full Marks: 120

- N.B. i) Answer any TWO questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is a smart grid? 03
- 1(b). Illustrate the architecture of the smart grid design. What are the main functions of smart grid components? 12
- 1(c). Provide an overview of the following technological requirements for a smart grid. 15
- (i) Information and communication technologies.
 - (ii) Sensing, measurement, control and automation technologies.
 - (iii) Power electronics and energy storage systems.
- 2(a). Using a diagram show a typical architecture for communication between a smart meter and the utility server. 15
- 2(b). Write down the application areas of the following communication technologies in smart grid: 15
- Bluetooth, ZigBee, WLAN, WPAN, WiMAX
- 3(a). Why load flow studies are essential in smart grid? Discuss the challenges of load flow in smart grid. 10
- 3(b). Provide an overview of the following performance analysis tools for smart grid design. 20
- (i) DSOPF, (ii) SSA, (iii) Congestion, and (iv) Contingency.

SECTION – B

- 4(a). What are the simplest thing and the complex thing about the smart grid? Write about major reasons for voltage stability problems in power system. 12
- 4(b). Consider a synchronous machine characterized by the parameters: $x_d = 1.0$, $x_q = 0.6$, $x'_d = 0.3$ per unit and negligible armature resistance. The machine is connected directly to an infinite bus of voltage 1.0 per unit. The generator is delivering a real power of 0.5 per unit at 0.8 power factor lagging. Determine the voltage behind transient reactive and the transient power-angle equation for the following cases: 10
- (i) Neglecting the saliency effect.
 - (ii) Including the effect of saliency.
- 4(c). What is the necessity of reactive power control in grid? Write the different methods of reactive power control. 08
- 5(a). What are the stability analysis tools for smart grid? Write the reasons of computational challenges in smart grids. 10
- 5(b). Classify the techniques of voltage stability assessment. Describe the VIPI method for voltage stability assessment. 10
- 5(c). How Artificial Neural Network (ANN) improves smart grid technology? Illustrate the Analytical Hierarchical Programming (AHP) method. 10
- 6(a). What is interoperability in smart grid environment? How cyber security is a threat to smart grids? 12
- 6(b). What is false data injection attacks in the smart grid? “Automatic Generation Control (AGC) is essential for power system”- Justify. 12
- 6(c). What are the main mitigation approaches to cyber security risk? 06

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2020

ESE 4125

(Safety and Environmental Aspects of Energy Projects)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | |
|--|----|
| 1(a). Define the followings: | 06 |
| (i) Hazard, (ii) Occupational accident, (iii) Occupational safety | |
| 1(b). What are the different methods of hazard identification? Briefly explain the HAZOP procedure. | 12 |
| 1(c). Briefly explain the following physical hazards: | 12 |
| (i) Frostbite, (ii) Heat stroke, (iii) Asphyxiation, and (iv) Ionizing radiation | |
| 2(a). What are the different methods of fire extinguishment? Define each method. | 06 |
| 2(b). What is meant by Fire Triangle? What are the factors that contribute to a fire incident? | 10 |
| 2(c). Illustrate the applications of fire fighting materials to combat different classes of fire. | 14 |
| 3(a). What are the limitations of Hazard Evaluation study? | 05 |
| 3(b). What are the common PPEs used in an industry? Note down their advantages. | 08 |
| 3(c). Classify the Emergency Preparedness Plan (EPP). | 05 |
| 3(d). What are the major causes of workplace accident? Explain the Man and Machine relationship of accident. | 12 |

SECTION – B

- | | |
|--|----|
| 4(a). Define: (i) Occupational health, (ii) Occupational diseases, and (iii) Occupational exposure, and (iv) Threshold Limit Value. | 10 |
| 4(b). How to provide the proper occupational health? Discuss. | 10 |
| 4(c). Write short notes on: (i) Urticaria, (ii) Pigmentary changes | 10 |
| 5(a). Explain the hazardous chemical dose-response relationship. | 06 |
| 5(b). Define the followings based on toxic material:
(i) Local effect, (ii) Systemic effect, (iii) Acute effect, and (iv) Chronic effect. | 08 |
| 5(c). What is meant by MSDS? List all the crucial contents of a MSDS. | 12 |
| 5(d). Write down the name of different types of hazardous chemical exposure process. | 04 |
| 6(a). Why is occupational ergonomics important? | 05 |
| 6(b). According to nature of disposal, classify the pollutants. | 05 |
| 6(c). Write short notes on: (Green house effect, (ii) SMOG, (iii) Acid rain. | 12 |
| 6(d). What are the purposes of (i) initial Environment Examination (IEE) and (ii) Environmental Impact Assessment (EIA) | 08 |

--- END ---